

Hidden deep sulfide cumulates beneath the Jinchuan Ni–Cu–platinum-group element deposit (China) inferred from Ni–Cu isotopes

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ABSTRACT

The mechanisms that concentrated metals in the world's single largest nickel–copper–platinum-group element (Ni–Cu–PGE) sulfide deposit, the Jinchuan deposit in northwestern China, remain enigmatic. Nickel and Cu isotopes are novel isotopic systems that may fill this knowledge gap as they are major constituents of sulfide melt, the main carrier of economically important metals. Sulfide ores in the Jinchuan deposit are characterized by systematically high $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values, which are positively correlated ($r^2 = 0.84$), and show well-defined correlations with PGEs, S, Ni, and Cu contents, and Ni/Pd, Cu/Pd, and Pd/Ir ratios. The isotopic signatures can be best explained by the early segregation of sulfide melt at depth prior to ascent of the parental magmas, storing isotopically light Ni and Cu in deep magma chambers. Subsequent sulfide melt segregation and fractionation in shallow portions of the magmatic conduit generated the currently explored sulfide ores, with heavy and positively correlated Ni–Cu isotopic signatures. Therefore, hidden deep orebodies are promising targets for further exploration beneath the Jinchuan deposit.

INTRODUCTION

Magmatic nickel–copper–platinum-group element (Ni–Cu–PGE) sulfide deposits account for ~56%, 3%, and >96% of the world's Ni, Cu, and PGE production, respectively, with giant deposits (e.g., Jinchuan in northwestern China, Norilsk in Siberia) containing the majority of the resources (Barnes and Lightfoot, 2005). World-class magmatic Ni–Cu–PGE sulfide camps typically comprise numerous deposits that are associated with komatiitic, basaltic, and mafic–ultramafic magmatism (Barnes and Lightfoot, 2005), including the Norilsk–Talnakh camp in Russia (23.2×10^6 Mt Ni and ~ 46.7 Mt Cu, $\sim 4 \times 10^6$ km³ magma; Ernst, 2014), the Sudbury impact structure in Canada (~ 19.8 Mt Ni

and ~ 17.0 Mt Cu, $\sim 3 \times 10^4$ km³ magma; Zieg and Marsh, 2005), and deposits in the North American Midcontinent Rift system (i.e., the Keweenaw large igneous province, $\sim 2 \times 10^6$ km³ magma; Cannon, 1992). The Jinchuan deposit is the world's single largest magmatic Ni–Cu–PGE sulfide deposit in terms of grade and tonnage (~ 6.25 Mt Ni and ~ 4.0 Mt Cu; Li and Ripley, 2011; Long et al., 2023), but it occurs as a single, small mafic–ultramafic conduit (~ 1 km³; Figs. S1 and S2 in the Supplemental Material¹), which formed from an estimated 3×10^2 km³ of magma. The processes that concentrated such a large amount of metals in a volumetrically small intrusion remain a topic of debate.

One of the main hindrances to characterizing the mechanisms for metal accumulation is the lack of geochemical and/or isotopic tools that can be used to track the behavior of sulfide melt. Nickel–Cu isotopes have been increasingly applied to the study of magmatic Ni–Cu sulfide

deposits over the past decade (Fig. 1; Gueguen et al., 2013; Hofmann et al., 2014; Malitch et al., 2014; Zhao et al., 2019, 2024; Brzozowski et al., 2021; Tang et al., 2020; Hiebert et al., 2022). Theoretical and experimental studies have demonstrated that sulfide melt that has equilibrated with silicate magmas is enriched in light Ni and Cu isotopes (Savage et al., 2015; Liu et al., 2018; Xia et al., 2019; Sheng et al., 2022). As a consequence, most sulfide ores in magmatic deposits worldwide are enriched in light Ni–Cu isotopes (Fig. 1). We demonstrate that the various types of sulfide ores in the Jinchuan deposit have distinctly heavy and positively correlated Ni–Cu isotopic compositions (Figs. 1–3; Fig. S3; $r^2 = 0.84$, $n = 27$). We argue that the unique Ni–Cu isotopic compositions of the Jinchuan deposit are indicative of a deep sulfide segregation event.

The sulfide ore samples from the Jinchuan deposit can be subdivided into three groups based on texture: disseminated, net-textured, and massive sulfide ores. Sulfides occur interstitial to silicate minerals in disseminated ores, as an interconnected network interstitial to silicates in net-textured ores, and as discordant veins in massive ores (Fig. S2). Based on the descriptions of Barnes and Lightfoot (2005), and petrological, mineralogical, and geochemical studies (Chen et al., 2013; Zhao et al., 2022a; Fig. S2), disseminated sulfides (<30 wt% sulfide) formed as a result of segregation of sulfide droplets, whereas massive (>50 wt% sulfide) and Cu-rich net-textured (30–50 wt% sulfide) ores formed via fractionation of accumulated sulfide melt. The samples that were chosen for Ni–Cu isotopic characterization exhibit mini-

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¹Supplemental Material. Detailed geological setting, analytical and calculating methods, and data sources for this study. Please visit <https://doi.org/10.1130/GEOL.S.27097324> to access the supplemental material; contact editing@geosociety.org with any questions.

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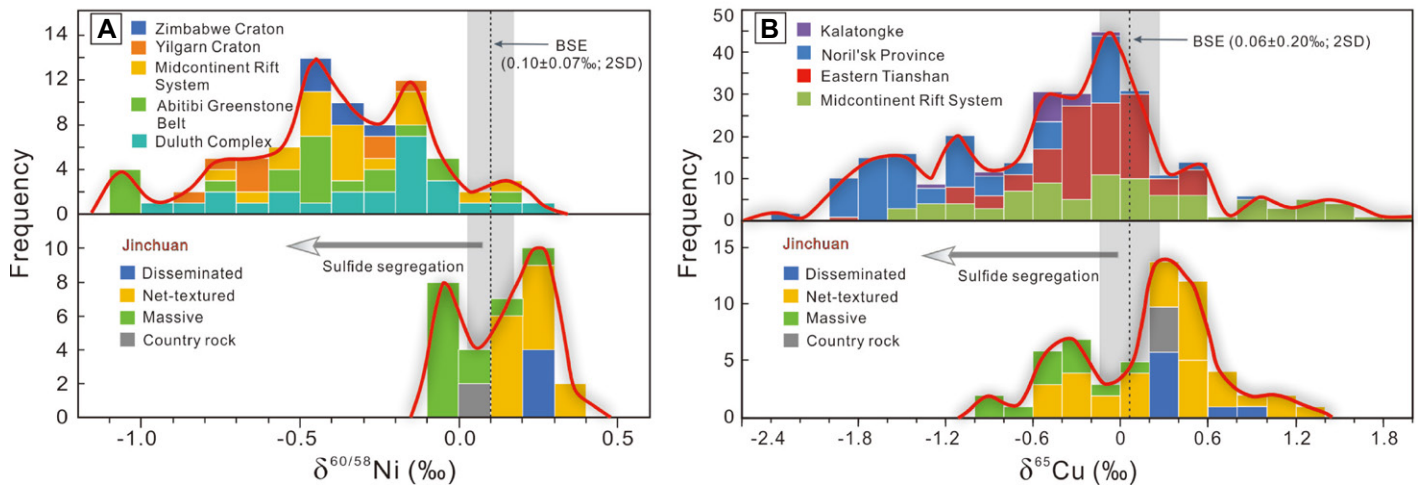


Figure 1. (A) Nickel isotopic compositions of bulk silicate earth (BSE; Klaver et al., 2020; Wang et al., 2021) and magmatic Ni-Cu sulfide deposits from the Zimbabwe Craton (Hofmann et al., 2014), the Yilgarn Craton (Western Australia; Gueguen et al., 2013), the North American Midcontinent Rift system (Smith et al., 2022), the Abitibi greenstone belt (Canada; Hiebert et al., 2022), the Duluth Complex (USA; Asp, 2006), as well as the Jinchuan Ni-Cu-platinum-group element (PGE) sulfide deposit in the North China Craton (Zhao et al., 2022a). (B) Copper isotopic compositions of BSE (Liu et al., 2015), and magmatic Ni-Cu sulfide clusters and deposits, including the Kalatongke mine (northwestern China; Tang et al., 2020); the Noril'sk Province (Russia; Malitch et al., 2014); those in the Eastern Tianshan (Zhao et al., 2019, 2022b; Tang et al., 2020), and the North American Midcontinent Rift system (Ripley et al., 2015; Smith et al., 2022); and the Jinchuan Ni-Cu-PGE sulfide deposit (Zhao et al., 2022a).

mal to no weathering or hydrothermal alteration, precluding these processes as controls on their isotopic compositions (Fig. S2). Furthermore, the $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values are negatively correlated with whole-rock PGE, S, Ni, and Cu contents (Fig. 2; Fig. S4), further ruling out the influence of secondary processes. The newly determined Ni isotopic compositions of both whole rocks and sulfide separates, coupled with previously reported Cu isotopic compositions

of chalcopyrite, provide new insights on the mechanisms that generated the Jinchuan deposit.

METHODS AND RESULTS

Detailed procedures for our Ni isotopic analysis are provided in the Supplemental Material. Whole-rock $\delta^{60/58}\text{Ni}$ values of disseminated ($+0.27\text{‰} \pm 0.01\text{‰}$, $n = 4$) and net-textured ($+0.22\text{‰} \pm 0.08\text{‰}$, $n = 6$) ores are systematically heavier than the $\delta^{60/58}\text{Ni}$ values of massive

sulfides ($-0.27\text{‰} \pm 0.03\text{‰}$, $n = 4$; Fig. 1). Chalcopyrite in disseminated and net-textured ores ($+0.36\text{‰} \pm 0.38\text{‰}$, $n = 42$) exhibit heavier $\delta^{65}\text{Cu}$ values than those in massive ores ($-0.44\text{‰} \pm 0.28\text{‰}$, $n = 11$; Zhao et al., 2022a). Additionally, the $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values are negatively correlated with whole-rock PGE, S, Ni, and Cu contents, as well as Ni/Pd and Cu/Pd ratios (Fig. 2; Fig. S4), but positively correlated with Pd/Ir ratios (Fig. 2).

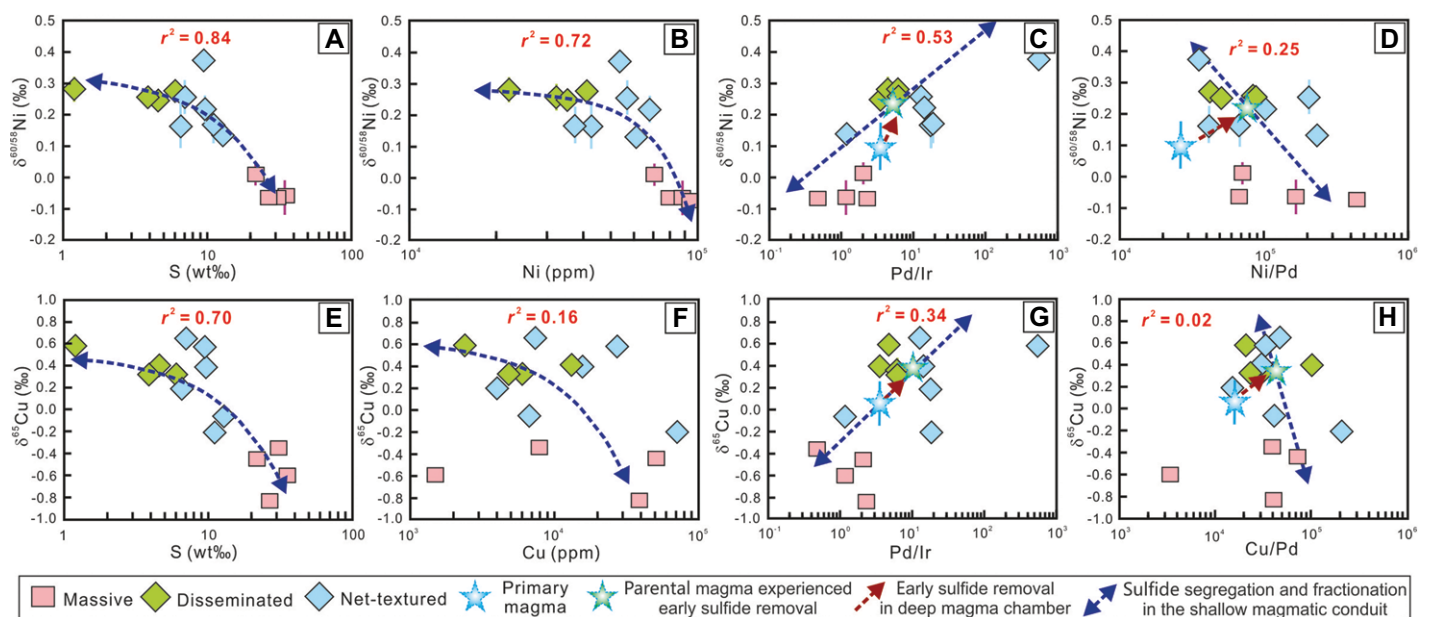


Figure 2. (A–D) Relationships between $\delta^{60/58}\text{Ni}$ and S content (A), Ni content (B), Pd/Ir ratio (C), and Cu/Pd ratio (D) in the Jinchuan Ni-Cu-platinum-group element (PGE) sulfide deposit in northwestern China. (E–H) $\delta^{65}\text{Cu}$ versus S content (E), Cu content (F), Pd/Ir ratio (G), and Pd/Os ratio (H) in the Jinchuan deposit. The Cu isotopic compositions, and S, Ni, Cu, Pd, and Ir contents are from Zhao et al. (2022a). **The primary magmas were assumed to have an ocean island basalt-like composition, with 7 ppb Pd, 173 ppm Ni, and 112 ppm Cu (Barnes and Lightfoot, 2005, and references therein).**

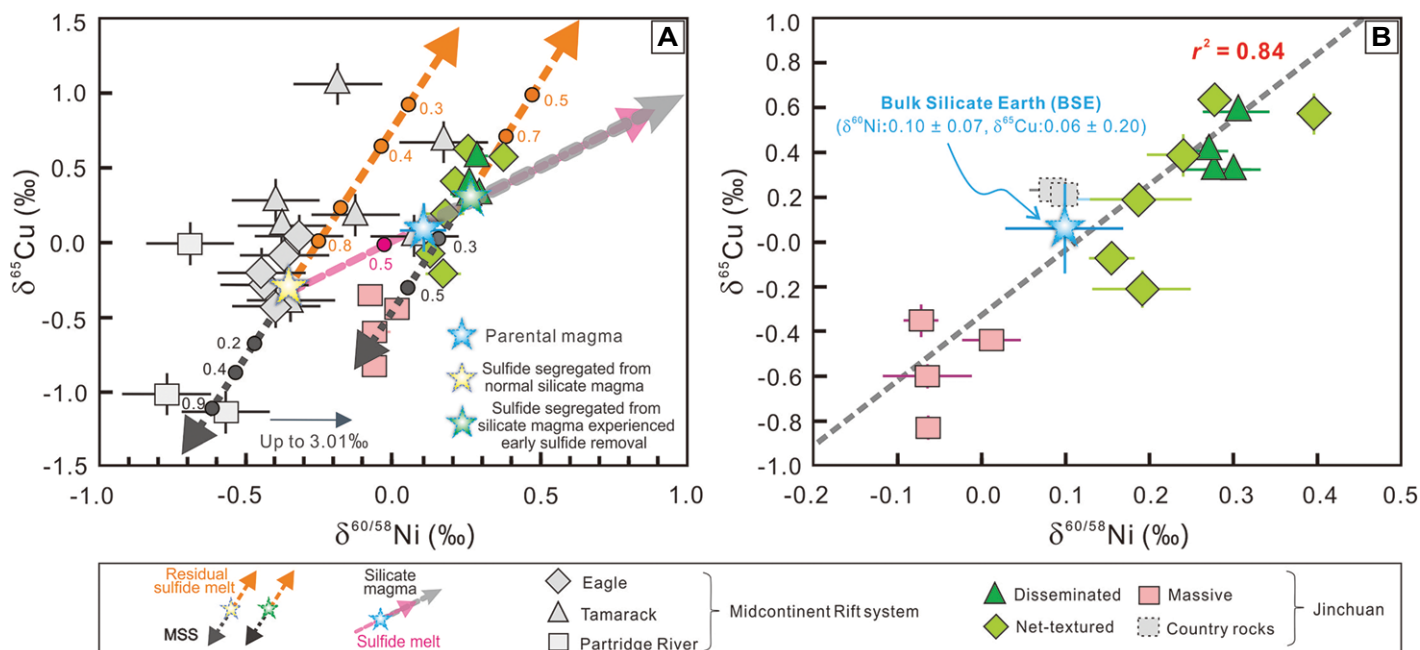


Figure 3. (A) Correlations between $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values for magmatic Ni–Cu sulfide deposits in the North American Midcontinent Rift system (Smith et al., 2022) and the Jinchuan Ni–Cu–platinum-group element (PGE) sulfide deposit in northwestern China. Details of the calculations are provided in the Supplemental Material (see text footnote 1). (B) Correlations between the $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values of whole-rock powders and pentlandite for different types of sulfide ores of the Jinchuan deposit. MSS—monosulfide solid solution. The $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ ranges of bulk silicate earth (BSE) are from Liu et al. (2015) and Wang et al. (2021).

DISCUSSION AND CONCLUSIONS

The Role of Sulfide Segregation and Sulfide Melt Fractionation

Despite possible inter-mineral isotope fractionation, the whole-rock Ni budget of sulfide ores at Jinchuan is dominated by pentlandite, which is supported by the statistically indistinguishable $\delta^{60/58}\text{Ni}$ values of whole-rock powders and pentlandite separates (Fig. S3A). Similarly, the whole-rock Cu budget is principally controlled by chalcopyrite given that it is the main Cu-bearing mineral in the deposit. Accordingly, we used whole-rock Ni isotope values and chalcopyrite Cu isotope values to track the mechanisms that generated the Jinchuan deposit.

In magmatic Ni–Cu–PGE systems, both Ni and Cu strongly partition into sulfide melt ($D_{\text{Cu}}^{\text{sulfide melt/silicate magma}} = 330 - 2130$, $D_{\text{Ni}}^{\text{sulfide melt/silicate magma}} = 250 - 1700$). Theoretical and experimental investigations and natural observations indicate that sulfide melt is enriched in light Ni–Cu isotopes relative to silicate melt (e.g., Savage et al., 2015; Xia et al., 2019; Liu et al., 2018; Chen et al., 2023; Yang et al., 2023). Furthermore, as sulfide melt evolves, the residual Cu-rich liquid becomes isotopically heavy (Zhao et al., 2019; Hiebert et al., 2022). Given that monosulfide solid solution (MSS) and intermediate solid solution (ISS) did not physically separate in disseminated sulfides, they are the best records of the Ni–Cu isotopic compositions of the sulfide melt that equilibrated with the silicate magmas. Variations in Ni–Cu isotopes during sulfide segregation and sulfide melt fractionation can be quantitatively

modeled using Rayleigh fractionation equations. The $\delta^{60/58}\text{Ni}$ ($+0.1\text{‰}$) and $\delta^{65}\text{Cu}$ ($+0.06\text{‰}$) values of the parental magmas are assumed to be similar to those of the bulk silicate earth (BSE; Liu et al., 2015; Klaver et al., 2020; Wang et al., 2021). The Ni (~ 350 ppm) and Cu (~ 120 ppm) concentrations of the parental magmas are from Chen et al. (2013). The Cu and Ni isotope fractionation factors between sulfide melt and silicate magma are both set at 0.99965 (Chen et al., 2013; Xia et al., 2019). The Cu and Ni isotope fractionation factors between MSS and sulfide melt are set at 0.999 (Zhao et al., 2019) and 0.9997 (Hiebert et al., 2022), respectively. Rayleigh fractionation modeling demonstrates that fractionation of sulfide melt that segregated from silicate magmas with BSE-like $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values can produce the Ni–Cu isotopic compositions of the Ni–Cu–PGE sulfide deposits in the Midcontinent Rift (Fig. 3A). The sulfide ores at Jinchuan exhibit a positive correlation between Ni and Cu isotope values, but are shifted toward heavier $\delta^{60/58}\text{Ni}$ values compared to deposits in the Midcontinent Rift (Fig. 3A). The elevated $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values of the disseminated sulfide ores at Jinchuan (Figs. 1 and 3) indicate that the parental magmas were isotopically heavy.

Origin of the Heavy Ni–Cu Isotopic Compositions

The heavy Ni–Cu isotopic compositions of disseminated ores at Jinchuan cannot be the result of mantle source heterogeneity or variations in mantle melting. Olivine and orthopyroxene contain more than 90% of the Ni budget of the

ambient mantle source, and both exhibit $\delta^{60/58}\text{Ni}$ values (olivine: $+0.07\text{‰} \pm 0.09\text{‰}$, 2 standard deviations [SD]; Klaver et al., 2020; Sheng et al., 2022; orthopyroxene: $+0.09\text{‰} \pm 0.08\text{‰}$, 2SD; Gall et al., 2017; Sheng et al., 2022) that are comparable to the BSE ($+0.11\text{‰} \pm 0.06\text{‰}$, 2SD; Klaver et al., 2020; Wang et al., 2021). Copper in mantle peridotites is mainly hosted by sulfides (Barnes and Lightfoot, 2005). Given that the Jinchuan intrusion crystallized from a magma derived from $\sim 18\%$ partial melting of mantle peridotite (Li and Ripley, 2011), the primary magmas are likely to have had mantle-like Ni–Cu isotopic compositions due to the exhaustion of sulfides and limited Ni–Cu isotope fractionation ($<0.15\text{‰}$) during melting of silicate minerals (Liu et al., 2015; Sheng et al., 2022).

Mantle metasomatism and melt-peridotite interaction can also be ruled out as having generated the heavy Ni–Cu isotopic compositions of the Jinchuan ores for the following reasons.

The Jinchuan intrusion crystallized from magmas derived from an enriched subcontinental lithospheric mantle (SCLM) source (Li and Ripley, 2011). However, metasomatic processes would have caused the mantle to exhibit a limited range of $\delta^{60/58}\text{Ni}$ values (e.g., Wang et al., 2021) and a large range of $\delta^{65}\text{Cu}$ values (e.g., Liu et al., 2015; Huang et al., 2017), but this is not observed. Refertilization of depleted SCLM by upwelling asthenospheric melts is not uncommon, leading to the precipitation of pyroxene $\pm$ spinel $\pm$ sulfide with isotopically light Ni–Cu isotopes (Wang and Becker, 2015). Consequently, it is unlikely that melting of refer-

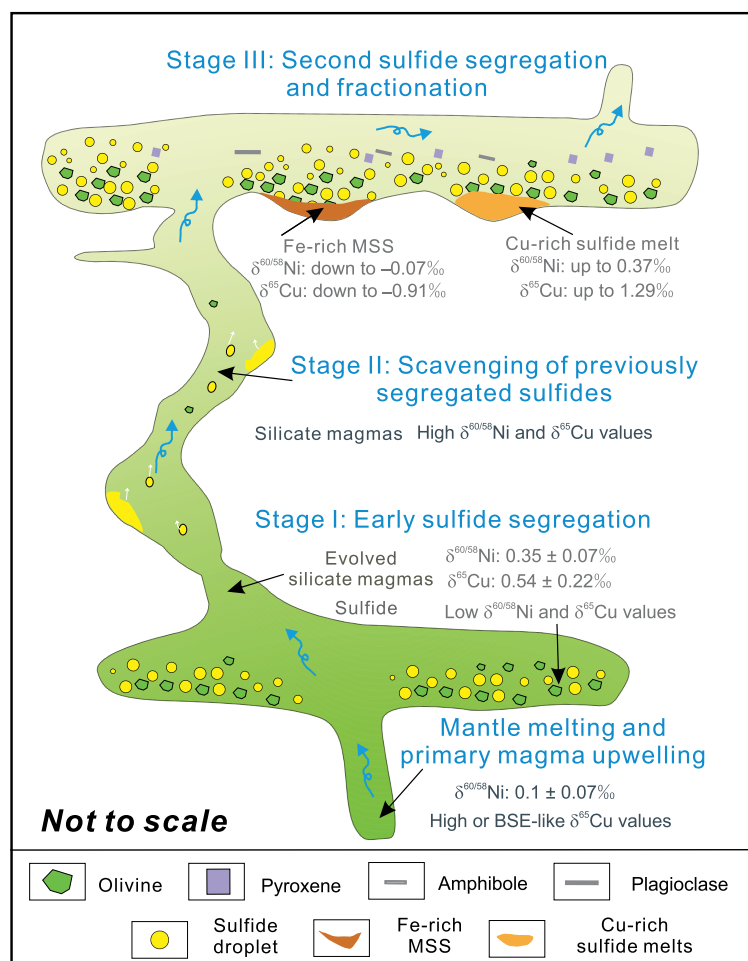


Figure 4. Schematic diagram illustrating Ni–Cu isotope variations in the Jinchuan Ni–Cu–platinum-group element (PGE) sulfide deposit in northwestern China. Stage I: Sulfide segregated in a deep magma chamber, leading to heavy Ni–Cu isotopes in the evolved magmas and later segregated sulfides. Stage II: New pulses of silicate magma scavenged previously segregated sulfides, adding isotopically heavy Ni and Cu to the magmas. Stage III: Sulfide segregation and fractionation further caused Ni–Cu isotope fractionation. MSS—monosulfide solid solution; BSE—bulk silicate earth.

tilized, sulfide-enriched SCLM would generate silicate magmas with both high $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values, as observed at Jinchuan. We simulated the effect of crustal contamination on the Ni–Cu isotopic compositions of the Jinchuan magma using a binary mixing model (Fig. S5). Within the contamination range of the Jinchuan deposit ($<30\%$; Tang et al., 2018), limited variations in $\delta^{60/58}\text{Ni}$ ($<0.1\text{‰}$) and $\delta^{65}\text{Cu}$ ($<0.4\text{‰}$) values are expected (Fig. S5).

The coupled enrichment of heavy Ni–Cu isotopes in the Jinchuan deposit can only be explained if substantial early sulfide removal occurred in the conduit system. In the Jinchuan deposit, disseminated sulfides formed from sulfide melt that experienced R factors (silicate magma/sulfide melt mass ratios) of ~ 100 – 3000 (e.g., Chen et al., 2013; Zhao et al., 2022a). High R factors would shift the Ni–Cu isotopic compositions of the sulfide melt toward the values of the host magma; the heavy Ni–Cu isotope values of the Jinchuan ores can, therefore, be obtained if the silicate magmas had heavy

$\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values (Fig. S6). Early sulfide segregation (up to 50%; Fig. 3A) prior to the silicate magmas being emplaced at shallow levels has also been inferred by spatial variations in PGE (Chen et al., 2023; Long et al., 2023), S/Se ratios, and Sr–Nd–S–C isotopes (Xue et al., 2023), and by the fact that the evolved silicate magmas became enriched in heavy Ni–Cu isotopes (stage I in Fig. 4 and Fig. S7). After initial sulfide saturation at depth, these evolved magmas became saturated in sulfide again as a result of the addition of crustal sulfur (Xue et al., 2023), segregating sulfides with heavy $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values that were scavenged by new pulses of silicate magma (stage II in Fig. 4). Subsequent segregation of sulfide melt from the evolved silicate magmas in the shallow portions of the magmatic conduit generated the Jinchuan disseminated sulfide ores, with heavy $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values (Fig. 3A; Fig. S8); sulfide melt–silicate magma fractionation further fractionated Ni–Cu isotopes (stage III in Fig. 4).

Implications for the Generation of and Exploration for Ni–Cu–PGE Deposits

It has generally been suggested that plume–rift intersections are ideal sites for the formation of magmatic sulfide deposits because of the development of large volumes of metal-bearing magma and the presence of well-defined conduits for transportation of this magma (Barnes and Lightfoot, 2005). Efficient scavenging of metals from silicate magmas by sulfide melts in shallow magmatic conduits is essential for the formation of magmatic Ni–Cu–PGE sulfide deposits (Barnes and Lightfoot, 2005). Sulfide saturation caused by the assimilation of crustal material in shallow magmatic conduits, and accumulation and fractionation of the sulfide melt, can generate the light Ni–Cu isotopic compositions that are typical of magmatic Ni–Cu–PGE sulfide deposits worldwide (Figs. 1 and 4; Hofmann et al., 2014; Ripley et al., 2015; Tang et al., 2021; Hiebert et al., 2022; Chen et al., 2023). In contrast, disseminated sulfides in the world-class Jinchuan deposit exhibit $\delta^{60/58}\text{Ni}$ and $\delta^{65}\text{Cu}$ values that are notably heavier than the BSE and other magmatic Ni–Cu–PGE sulfide deposits (Figs. 1 and 4). These heavy Ni–Cu isotopic signatures suggest that sulfide melt was removed from the parental magma during an earlier segregation event at depth, generating isotopically heavy values in the evolved magma. A second sulfide segregation event and subsequent sulfide melt fractionation at shallower levels generated the sulfide ores with heavy and positively correlated Ni–Cu isotopes (Figs. 3 and 4). In this scenario, hidden orebodies with light Ni–Cu isotopic compositions are promising targets for further exploration in the Jinchuan deposit. Indeed, an orebody that extends downward was recently discovered during underground prospecting at Jinchuan (Long et al., 2023). Thus, enrichment of heavy Ni–Cu isotopes in disseminated sulfides can be regarded as an effective tracer for hidden orebodies. Furthermore, multiple episodes of sulfide segregation, scavenged by new pulses of silicate magma, can generate metal-rich parental magmas that fuel the formation of giant magmatic Ni–Cu–PGE sulfide deposits, especially in volumetrically small mafic-ultramafic intrusions.

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